## 2.3 Moment of inertia of thin plates

Having learnt the MI of plane area, let us now look in to the MI of thin laminae (plates/discs) having some mass. The MI of a material body (mass moment of inertia) is defined by

$I_{xx} = \int y^2 \, dm$

$I_{yy} = \int x^2 \, dm$ , where $dm$ is the mass of the element.

Consider a thin plate (lamina) of uniform thickness t and mass m. Consider a small elemental block of mass dm at a distance (x,y) from the co-ordinate axes.

$dm = dA.t.\rho$ where ρ is the density of the material and dA is the cross-sectional area of the element.

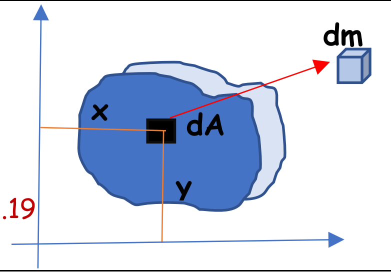


Fig. 2.19

By definition $I_{xx} = \int y^2 \, dm = \int y^2 \, t.\rho.dA = t.\rho \int y^2 \, dA$

= $t.\rho$. (moment of inertia of the cross-sectional area with respect to the x-axis)

$$I_{yy} = \int x^2 \, dm = \int x^2 \, t.\rho.dA = t.\rho \int x^2 \, dA$$

= $t.\rho$. (moment of inertia of the cross-sectional area with respect to the y-axis)

Moment of inertia of a lamina (a three-dimensional body with a small uniform thickness) about an axis can be found out by multiplying the moment of inertia of the cross-sectional area about the same axis by ($t\rho$)

Q1. Determine the MI of a circular lamina of mass m and radius r about the centroidal x, y, z axes shown.

Moment of inertia of a circular lamina with respect to the x-axis passing through the centroid = $t.\rho$. (moment of inertia of the circular area with respect to the x-axis)

$$I_{xx} = t.\rho \left(\frac{\pi r^4}{4}\right) = \pi r^2.t.\rho \,.\frac{r^2}{4} = \frac{mr^2}{4}$$

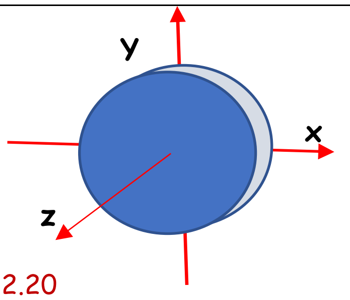


Fig. 2.20

where $\pi r^2$ is the area of the circle, $t$ is the thickness of the lamina, and $\rho$ is the density of the material so that $\pi r^2 t.\rho = m$.

Similarly,

$$I_{yy} = t.\rho \left(\frac{\pi r^4}{4}\right) = \pi r^2 t.\rho \,.\frac{r^2}{4} = \frac{mr^2}{4}$$

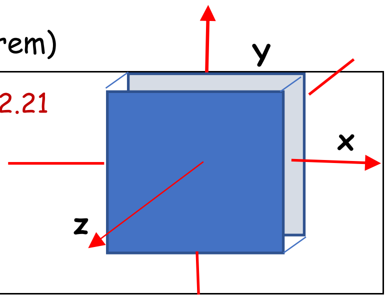
The polar moment of inertia of the circular disc (MI of the disc with respect to the z-axis perpendicular to the plane of the lamina) = $t.\rho$. (moment of inertia of the circular area with respect to the z-axis)

$$I_{zz} = t.\rho \left(\frac{\pi r^4}{2}\right) = \pi r^2 t.\rho \,.\frac{r^2}{2} = \frac{mr^2}{2}$$

You can see that $I_{zz} = I_{xx} + I_{yy}$ (Obeys the perpendicular axis theorem)

Q2. Determine the MI of a rectangular lamina of mass m with sides b,h with respect to the centroidal x, y, z axes.

Moment of inertia of a rectangular lamina with respect to the x-axis passing through the centre = $t.\rho$. (moment of inertia of the rectangular area with respect to the x-axis)

Fig. 2.21


$$I_{xx} = t.\rho\left(\frac{bh^3}{12}\right) = bh.t.\rho.\frac{h^2}{12} = \frac{mh^2}{12}$$

where $bh$ is the area of the rectangle, $t$ is the thickness of the lamina, and $\rho$ is the density of the material so that $bh.t.\rho = m$.

$$I_{yy} = t.\rho\left(\frac{b^3h}{12}\right) = bh.t.\rho.\frac{b^2}{12} = \frac{mb^2}{12}, \qquad and \qquad I_{zz} = I_{xx} + I_{yy} = \frac{m(b^2+h^2)}{12}$$

Q 3. Determine the MI of a triangular lamina of mass m with sides b,h with respect to the centroidal x, y, z axes.

Moment of inertia of a triangular lamina with respect to the x-axis passing through the centroid = $t.\rho$. (moment of inertia of the triangular area with respect to the x-axis)

$$I_{xx} = t.\rho\left(\frac{bh^3}{36}\right) = \frac{1}{2}bh.t.\rho.\frac{h^2}{18} = \frac{mh^2}{12}$$

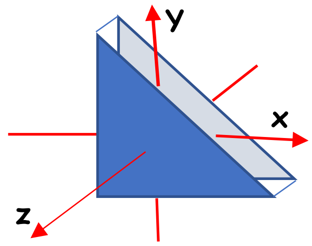


Fig. 2.22

where $\frac{1}{2}bh$ is the area of the triangle, $t$ is the thickness of the lamina, and $\rho$ is the density of the material so that $\frac{1}{2}bh.t.\rho = m$.

$$I_{yy} = t.\rho\left(\frac{b^3h}{36}\right) = \frac{1}{2}bh.t.\rho.\frac{b^2}{18} = \frac{mb^2}{18}, \qquad and \qquad I_{zz} = I_{xx} + I_{yy} = \frac{m(b^2+h^2)}{18}$$

#Exercise

State and prove the parallel axis theorem for mass moment of inertia and hence determine the moment of inertia of the rectangular and triangular laminae (Q2, Q3) with respect to the coordinate axes passing through the left-bottom corner.

## 2.4 Moment of inertia of material bodies

Now we can find out the moment of inertia of some of material bodies like cylinder, cone, sphere etc.

Q 1. Determine the MI of a cylinder of mass m, radius r, and length l with respect to the centroidal axes

MI of a cylinder about a horizontal axis passing through its c.g ($I_{xx}$ or $I_{zz}$)

Divide the cylinder into elementary circular discs and consider one such disc of thickness $dy$ at a distance $y$ from the x-axis.

Mass of this elemental disc, $dm = \pi r^2.dy.\rho$

Moment of inertia of a circular lamina about an axis passing through its centroid and parallel to the plane of the lamina $= \frac{mr^2}{4}$

Therefore, MI of the elemental disc about the horizontal axis passing through its centre $= \frac{dm\, r^2}{4} = \frac{\pi r^2.dy.\rho\, r^2}{4}$

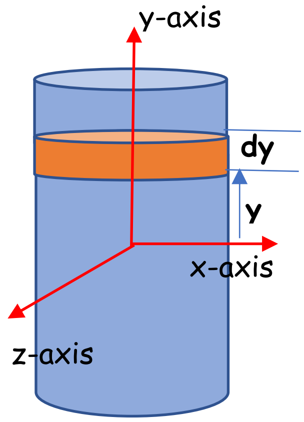


Fig. 2.23a

By applying the parallel axis theorem, we can find out the MI of the elemental disc with respect the x-axis shown in figure, which is = MI of the elemental disc about the horizontal axis passing through its centre + (mass of the element x square of the distance between the parallel axes)

$$dI_{xx} = \frac{dm.\ r^2}{4} + dm.y^2 = \frac{\pi r^2.dy.\rho\, r^2}{4} + \pi r^2.dy.\rho\, y^2$$

Therefore, the total moment of inertia of the whole cylinder about the x-axis

$$I_{xx} = \int dI_{xx} = \int_{-\frac{l}{2}}^{\frac{l}{2}} \left( \frac{\pi r^2 \,.dy.\rho\, r^2}{4} + \pi r^2 \,.dy.\rho\, y^2 \right)$$

The integration is with respect to variable y. To cover the whole cylinder, integration should extend from $y = -\frac{l}{2}$ to $y = \frac{l}{2}$

$$I_{xx} = \frac{\pi r^2 \,.\rho\, r^2}{4} \int_{-\frac{l}{2}}^{\frac{l}{2}} dy + \pi r^2 \,.\rho \int_{-\frac{l}{2}}^{\frac{l}{2}} y^2\, dy. = \frac{\pi r^2\, \rho\, r^2}{4} l + \pi r^2 \,.\rho \frac{2 l^3}{3 \times 8} = \pi r^2\, l\, \rho \left( \frac{r^2}{4} + \frac{l^2}{12} \right)$$

$$I_{xx} = m \left( \frac{r^2}{4} + \frac{l^2}{12} \right) \; kg.m^2$$

Since the distribution of mass of cylinder is the same with respect to x and z axis

$$I_{zz} = m \left( \frac{r^2}{4} + \frac{l^2}{12} \right) \; kg.m^2$$

The polar moment of inertia of the cylinder with respect to the y-axis (geometrical axis) can be obtained by integrating the polar moment of inertia of the elemental disc about the same axis.

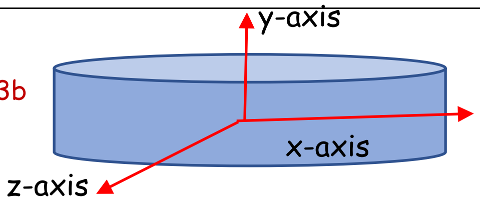
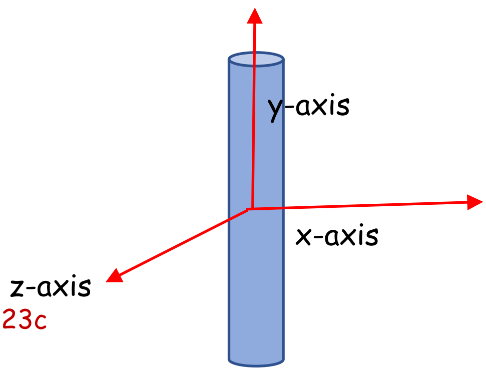
$$dI_{yy} = \frac{dm.\ r^2}{2} = \frac{\pi r^2 \,.dy.\rho\, r^2}{2}$$

$$I_{yy} = \int dI_{yy} = \int_{-\frac{l}{2}}^{\frac{l}{2}} \left( \frac{\pi r^2 \,.dy.\rho\, r^2}{2} \right) = \frac{\pi r^2\, \rho\, r^2}{2} l = \pi r^2 l\, \rho\, \frac{r^2}{2}$$

$$I_{yy} = \frac{m\, r^2}{2} \quad kg.m^2$$

The polar moment of inertia of the cylinder about its geometrical axis is not influenced by its length. With respect to its geometrical axis, the cylinder behaves as same as a circular disc.

Please note that the perpendicular axis theorem is not valid for three-dimensional material bodies.

| | |
|---|---|
| If $r \gg l$, then $\frac{l^2}{12}$ can be neglected.<br>Then $I_{xx} = I_{zz} = \frac{m\, r^2}{4}$ (same as circular disc) | Fig. 2.23b   |
| If $l \gg r$, then $\frac{r^2}{4}$ can be neglected.<br>Then $I_{xx} = I_{zz} = \frac{m\, l^2}{12}$ (Slender bar) |   Fig. 2.23c |

**Q 2. Determine the MI of a sphere of mass m and radius R about its centroidal axes.**

MI of a cylinder about any of its diameter ($I_{xx}$ or $I_{yy}$ or $I_{zz}$)

Divide the sphere into elementary circular discs and consider one such disc of radius r and thickness $dy$ at a distance $y$ from the x-axis.

Mass of this elemental disc, $dm = \pi r^2 \,.dy.\rho$

From the right triangle, $r^2 = R^2 - y^2$

$$dm = \pi(R^2 - y^2)\,.dy.\rho$$

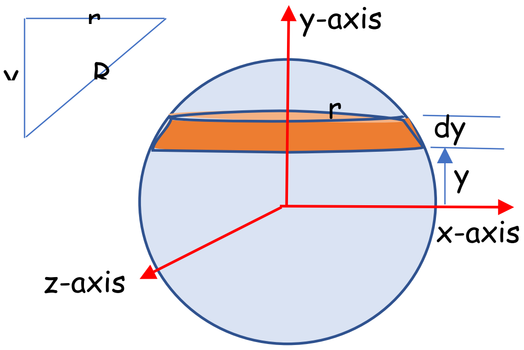


Fig. 2.24

Moment of inertia of a circular lamina about an axis passing through its centroid and parallel to the plane of the lamina $= \frac{mr^2}{4}$

Therefore, MI of the elemental disc about the horizontal axis passing through its centre $= \frac{dm\,r^2}{4}$

By applying the parallel axis theorem, we can find out the MI of the elemental disc with respect the x-axis shown in figure, which is = MI of the elemental disc about the horizontal axis passing through its centre + (mass of the element x square of the distance between the parallel axes)

$$dI_{xx} = \frac{dm.\;r^2}{4} + dm.y^2 = \frac{\pi(R^2 - y^2)\,.dy.\rho\,(R^2 - y^2)}{4} + \pi(R^2 - y^2)\,.dy.\rho\,y^2$$

Therefore, the total moment of inertia of the whole cylinder about the x-axis

$$I_{xx} = \int dI_{xx} = \int_{-R}^{R}\left(\frac{\pi(R^2 - y^2)\,.dy.\rho\,(R^2 - y^2)}{4} + \pi(R^2 - y^2)\,.dy.\rho\,y^2\right)$$

The integration is with respect to variable y. To cover the whole sphere, integration should extend from $y = -R$ to $y = R$

$$I_{xx} = \frac{\pi\,\rho}{4}\int_{-R}^{R}(R^4 - 2R^2y^2 + y^4)\,.dy + \pi\,\rho\int_{-R}^{R}(R^2\,y^2 - y^4)\,dy$$

$$= \frac{\pi\,\rho}{4}(2R^5 - \frac{4R^5}{3} + \frac{2R^5}{5}) + \pi\,\rho\left(\frac{2R^5}{3} - \frac{2R^5}{5}\right)$$

$$= \frac{\pi\,\rho}{4}\left(\frac{16R^5}{15}\right) + \pi\,\rho\left(\frac{4R^5}{15}\right) = \pi\,\rho\left(\frac{8R^5}{15}\right) = \left(\frac{4\pi\,R^3}{3}\rho\right)\left(\frac{2R^2}{5}\right)$$

$$I_{xx} = I_{yy} = I_{zz} = \frac{2}{5}mR^2 \quad kg.m^2$$

**Q 3. Determine the MI of a cone of mass m, radius R, and height h with respect to the axes shown.**

MI of a cone about any of its base diameter ($I_{xx}$ or $I_{zz}$)

Divide the cone into elementary circular discs and consider one such disc of radius r and thickness $dy$ at a distance $y$ from the x-axis.

Mass of this elemental disc, $dm = \pi r^2 \,.dy.\rho$

From the similar triangles,

$$\frac{r}{R} = \frac{h-y}{h} \quad \ggg r = \frac{R}{h}(h-y)$$

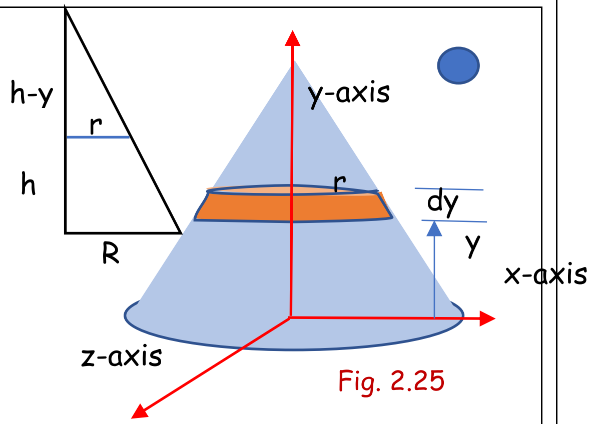


Fig. 2.25

$$dm = \pi \frac{R^2}{h^2}(h-y)^2 \,.dy.\rho$$

Moment of inertia of a circular lamina about an axis passing through its centroid and parallel to the plane of the lamina $= \frac{mr^2}{4}$

Therefore, MI of the elemental disc about the horizontal axis passing through its centre $= \frac{dm\, r^2}{4}$

By applying the parallel axis theorem, we can find out the MI of the elemental disc with respect the x-axis shown in figure, which is = MI of the elemental disc about the horizontal axis passing through its centre + (mass of the element x square of the distance between the parallel axes)

$$dI_{xx} = \frac{dm.\ r^2}{4} + dm.y^2 = \frac{\pi \frac{R^2}{h^2}(h-y)^2\,.dy.\rho.\frac{R^2}{h^2}(h-y)^2}{4} + \pi \frac{R^2}{h^2}(h-y)^2\,.dy.\rho\, y^2$$

Therefore, the total moment of inertia of the whole cylinder about the x-axis

$$I_{xx} = \int dI_{xx} = \int_0^h \left( \frac{\pi\,\rho\,\frac{R^4}{h^4}(h-y)^4}{4} dy + \pi\,\rho \frac{R^2}{h^2}(h-y)^2\, y^2 dy \right)$$

The integration is with respect to variable y. To cover the whole cone, integration should extend from $y = 0$ to $y = h$

$$I_{xx} = \frac{\pi\,\rho}{4}\frac{R^4}{h^4}\int_0^h (h-y)^4 dy + \pi\,\rho \frac{R^2}{h^2}\int_0^h (h^2\,y^2 - 2hy^3 + y^4)\,dy$$

$$= \frac{\pi\,\rho}{4}\frac{R^4}{h^4}\int_h^0 -(t)^4 dt + \pi\,\rho \frac{R^2}{h^2}\left(\frac{h^5}{3} - \frac{2h^5}{4} + \frac{h^5}{5}\right)$$

$$= \frac{\pi\,\rho}{4}\frac{R^4}{h^4}\left(\frac{h^5}{5}\right) + \pi\,\rho \frac{R^2}{h^2}\left(\frac{h^5}{30}\right) = \left(\frac{1}{3}\pi\,R^2 h\,\rho\right)\left(\frac{3}{20}R^2 + \frac{1}{10}h^2\right)$$

$$I_{xx} = I_{zz} = \frac{1}{20} m(3R^2 + 2h^2) \quad kg.m^2$$

#Exercise 1

Calculate the moment of inertia of the cone with respect to its geometrical axis